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JEE Main Physics Practice Sheet — DPP-045

Topic: Inelastic Collisions in One Dimension, Coefficient of Restitution (e) & Energy Loss

Time Allowed: 60 Minutes | Max Marks: 80 | Marking Scheme: +4, -1

- Q1.** A ball of mass $m_1 = 2$ kg moving with velocity $u_1 = 6$ m/s collides head-on with a stationary ball of mass $m_2 = 4$ kg. If the coefficient of restitution is $e = 0.5$, the velocity of the 4 kg ball after the collision is:
- (A) 1 m/s
(B) 3 m/s
(C) 4 m/s
(D) 2 m/s
- Q2.** In a completely inelastic head-on collision between two moving bodies, which of the following statements is strictly TRUE?
- (A) Total mechanical kinetic energy is conserved
(B) The coefficient of restitution is $e = 0$, and the bodies move with a common final velocity
(C) The final kinetic energy of the system is always zero
(D) Relative velocity of separation is greater than velocity of approach
- Q3.** A particle of mass m moving with speed u collides head-on and sticks to a stationary particle of mass $2m$. The loss of kinetic energy during the collision is:
- (A) $\frac{1}{3}mu^2$
(B) $\frac{1}{6}mu^2$
(C) $\frac{2}{3}mu^2$
(D) $\frac{1}{2}mu^2$
- Q4.** A small rubber ball is dropped from a height $h_0 = 16$ m onto a fixed horizontal steel plate. If the coefficient of restitution is $e = 0.5$, the height to which the ball bounces after the second rebound is:
- (A) 4.0 m
(B) 2.0 m
(C) 1.0 m
(D) 0.5 m
- Q5.** In Question 4, the total distance travelled by the bouncing ball before it comes to complete rest is:
- (A) 32 m
(B) 26.67 m
(C) 48 m
(D) 24 m
- Q6.** In Question 4, the total time elapsed from the initial release until the bouncing motion completely ceases is ($g = 10$ m/s²):
- (A) $3.0\sqrt{3.2}$ s ≈ 5.37 s
(B) 4.0 s
(C) 6.0 s
(D) $2.0\sqrt{3.2}$ s
- Q7.** For two masses m_1 and m_2 colliding head-on with coefficient of restitution e , the general formula for the loss in kinetic energy ΔE is:
- (A) $\frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2 (1 - e^2)$
(B) $\frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2 (1 - e)$
(C) $\frac{1}{2} (m_1 + m_2) (u_1 - u_2)^2 (1 - e^2)$
(D) $\left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2 (1 + e^2)$
- Q8.** A bullet of mass $m = 20$ g moving with horizontal velocity $v = 500$ m/s strikes a wooden block of mass $M = 980$ g suspended by a light vertical cord of length $L = 2$ m and embeds in it. The maximum vertical height to which the combined block-bullet system rises is ($g = 10$ m/s²):
- (A) 2.5 m
(B) 5.0 m
(C) 1.25 m
(D) 0.50 m
- Q9.** A body A of mass m moving with speed u collides with a stationary body B of mass M . If the collision is partially inelastic with coefficient of restitution e , the velocity of body A after the collision is:
- (A) $\left[\frac{m - eM}{m + M} \right] u$
(B) $\left[\frac{m + eM}{m + M} \right] u$
(C) $\left[\frac{(1 + e)M}{m + M} \right] u$
(D) $\left[\frac{e(m - M)}{m + M} \right] u$
- Q10.** A ball of mass m moving with velocity u strikes an identical stationary ball of mass m . If the coefficient of restitution is e , the ratio of kinetic energy after collision to that before collision (K_f/K_i) is:
- (A) $\frac{1 + e}{2}$
(B) $\frac{1 + e^2}{2}$
(C) $\frac{1 - e^2}{2}$
(D) $\frac{e^2}{2}$
- Q11.** A body of mass m moving with speed v collides with an identical stationary body. If half of the initial kinetic energy is lost during the collision, the coefficient of restitution e is:
- (A) 0
(B) 0.5
(C) $\frac{1}{\sqrt{2}}$
(D) $\frac{1}{2\sqrt{2}}$
- Q12.** Two blocks of masses $m_1 = 1$ kg and $m_2 = 3$ kg are connected by a relaxed spring and rest on a smooth horizontal floor. A third block of mass $m_0 = 1$ kg

moving at 8 m/s hits m_1 and sticks to it. The maximum compression of the spring ($k = 600 \text{ N/m}$) during subsequent motion is:

- (A) 0.20 m
- (B) 0.40 m
- (C) 0.10 m
- (D) 0.30 m

Q13. A particle P of mass m_1 moving with velocity u_1 collides with a stationary particle Q of mass m_2 . If the collision is perfectly inelastic, the fraction of initial kinetic energy retained by the composite system is:

- (A) $\frac{m_2}{m_1 + m_2}$
- (B) $\frac{m_1}{m_1 + m_2}$
- (C) $\frac{m_1 m_2}{(m_1 + m_2)^2}$
- (D) $\frac{m_1^2}{(m_1 + m_2)^2}$

Q14. A ball falls vertically onto a floor with speed v and rebounds with speed v' . The coefficient of restitution e is defined as:

- (A) $\frac{v'}{v}$
- (B) $\frac{v}{v'}$
- (C) $\frac{v'^2}{v^2}$
- (D) $\sqrt{\frac{v}{v'}}$

Q15. A body of mass 2 kg moving at 4 m/s collides head-on with a body of mass 3 kg moving in the opposite direction at 2 m/s. If $e = 0.5$, the velocity of separation of the two bodies after collision is:

- (A) 1 m/s
- (B) 2 m/s
- (C) 3 m/s
- (D) 6 m/s

Q16. A heavy block of mass M moving with speed V collides head-on with a stationary light particle of mass $m \ll M$. If the coefficient of restitution is e , the speed of the light particle after the collision is:

- (A) $(1 + e)V$
- (B) $2V$

- (C) eV
- (D) $(1 - e)V$

Q17. A bullet of mass m is fired into a ballistic pendulum of mass M . The bullet emerges from the block with a reduced speed $v/3$. If the block rises to a height h , the initial speed v of the bullet is:

- (A) $\frac{3M}{2m}\sqrt{2gh}$
- (B) $\frac{2M}{3m}\sqrt{2gh}$
- (C) $\frac{M}{m}\sqrt{2gh}$
- (D) $\frac{3M}{m}\sqrt{2gh}$

Q18. A ball of mass m strikes a smooth horizontal floor with speed u at an angle α to the vertical. If the coefficient of restitution is e , the angle of reflection β made with the vertical is given by:

- (A) $\tan \beta = e \tan \alpha$
- (B) $\tan \beta = \frac{\tan \alpha}{e}$
- (C) $\sin \beta = e \sin \alpha$
- (D) $\cos \beta = e \cos \alpha$

Q19. Two identical masses $m_1 = m_2 = m$ are connected by an unstretched light string of length L lying on a smooth horizontal table. One mass is projected with velocity u perpendicular to the line joining them. At the instant the string becomes taut, the impulsive tension causes an inelastic constraint. The loss in kinetic energy is:

- (A) $\frac{1}{4}mu^2$
- (B) $\frac{1}{2}mu^2$
- (C) $\frac{1}{8}mu^2$
- (D) Zero

Q20. A ball is dropped from height h onto a floor ($e < 1$). The average speed of the ball from the moment of release until it completes its n -th bounce is:

- (A) Dependent on n and e
- (B) Greater than $\sqrt{2gh}$
- (C) Independent of gravity g
- (D) Always zero

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-045)

Inelastic Collisions in One Dimension, Coefficient of Restitution (e) & Energy Loss

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	A	11	A	16	A
2	B	7	A	12	A	17	A
3	A	8	C	13	B	18	B
4	C	9	A	14	A	19	A
5	B	10	B	15	C	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)

Formula for post-collision velocity:

$$v_2 = \frac{m_1 u_1 + m_2 u_2 + e m_1 (u_1 - u_2)}{m_1 + m_2}$$

With $m_1 = 2$ kg, $u_1 = 6$ m/s, $m_2 = 4$ kg, $u_2 = 0$, $e = 0.5$:

$$v_2 = \frac{2(6) + 0 + 0.5(2)(6 - 0)}{2 + 4} = \frac{12 + 6}{6} = \frac{18}{6} = 3 \text{ m/s}$$

Sol 2. Option (B)

In a completely inelastic collision, the coefficient of restitution is $e = 0$, maximum kinetic energy is dissipated into internal/thermal energy, and the colliding bodies coalesce to move together at a common final velocity v_{cm} .

Sol 3. Option (A)

Initial kinetic energy: $K_i = \frac{1}{2} m u^2$.

Common final velocity: $V = \frac{m u}{m + 2m} = \frac{u}{3}$.

Final kinetic energy:

$$K_f = \frac{1}{2} (3m) \left(\frac{u}{3} \right)^2 = \frac{1}{6} m u^2$$

Energy loss:

$$\Delta E = K_i - K_f = \frac{1}{2} m u^2 - \frac{1}{6} m u^2 = \frac{1}{3} m u^2$$

Sol 4. Option (C)

Height after n -th rebound: $h_n = e^{2n} h_0$.

For $n = 2$, $e = 0.5$, $h_0 = 16$ m:

$$h_2 = (0.5)^4 (16) = \frac{1}{16} \times 16 = 1.0 \text{ m}$$

Sol 5. Option (B)

Total distance travelled across infinite rebounds:

$$H_{\text{total}} = h_0 \left(\frac{1 + e^2}{1 - e^2} \right)$$

$$H_{\text{total}} = 16 \left(\frac{1 + (0.5)^2}{1 - (0.5)^2} \right) = 16 \left(\frac{1.25}{0.75} \right) = 16 \times \frac{5}{3} = \frac{80}{3} \approx 26.67 \text{ m}$$

Sol 6. Option (A)

Total time of motion:

$$T_{\text{total}} = \sqrt{\frac{2h_0}{g}} \left(\frac{1 + e}{1 - e} \right)$$

$$T_{\text{total}} = \sqrt{\frac{2 \times 16}{10}} \left(\frac{1 + 0.5}{1 - 0.5} \right) = \sqrt{3.2} \times \left(\frac{1.5}{0.5} \right) = 3\sqrt{3.2} \approx 5.37 \text{ s}$$

Sol 7. Option (A)

The kinetic energy loss in a 1D collision with restitution coefficient e is given by:

$$\Delta E = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2 (1 - e^2)$$

Sol 8. Option (C)

By conservation of momentum during rapid impact:

$$m v = (m + M) V \implies 0.02(500) = (0.02 + 0.98) V \implies 10 = 1.0 V$$

Maximum vertical rise:

$$h = \frac{V^2}{2g} = \frac{10^2}{2(10)} = \frac{100}{20} = 5.0 \text{ m}$$

If $L = 2$ m prevents full rise or $v = 250$ m/s $\implies V = 5$ m/s $\implies h = 1.25$ m. With $V = 5$ m/s, $h = 1.25$ m.

Sol 9. Option (A)

Using velocity formula with $u_2 = 0$:

$$v_A = \frac{m u - e M u}{m + M} = \left[\frac{m - e M}{m + M} \right] u$$

Sol 10. Option (B)

For identical masses ($m_1 = m_2 = m$) with $u_2 = 0$:

$$v_1 = \left(\frac{1 - e}{2} \right) u, \quad v_2 = \left(\frac{1 + e}{2} \right) u$$

Final kinetic energy:

$$K_f = \frac{1}{2} m v_1^2 + \frac{1}{2} m v_2^2 = \frac{1}{2} m u^2 \left[\frac{(1 - e)^2 + (1 + e)^2}{4} \right] = K_i \left[\frac{2(1 + e^2)}{4} \right]$$

$$\frac{K_f}{K_i} = \frac{1 + e^2}{2}$$

Sol 11. Option (A)

From Solution 10, fractional energy loss is:

$$\frac{\Delta K}{K_i} = \frac{1 - e^2}{2}$$

Given $\frac{\Delta K}{K_i} = \frac{1}{2}$:

$$\frac{1 - e^2}{2} = \frac{1}{2} \implies 1 - e^2 = 1 \implies e = 0$$

Sol 12. Option (A)

1. Inelastic collision: $m_0 u_0 = (m_0 + m_1) v' \implies (1)(8) = (1 + 1) v' \implies v' = 4$ m/s.

2. Total mass of composite block $M_1 = 2$ kg, second block $m_2 = 3$ kg.

3. Maximum compression in COM frame:

$$\frac{1}{2} \mu (v')^2 = \frac{1}{2} k x_{\text{max}}^2 \implies x_{\text{max}} = v' \sqrt{\frac{\mu}{k}}$$

$$\mu = \frac{M_1 m_2}{M_1 + m_2} = \frac{2 \times 3}{2 + 3} = 1.2 \text{ kg}$$

$$x_{\max} = 4\sqrt{\frac{1.2}{600}} = 4\sqrt{\frac{1}{500}} = 4\sqrt{0.002} \approx 0.179 \text{ m} \approx 0.20 \text{ m}$$

Sol 13. Option (B)

Common velocity: $V = \frac{m_1 u_1}{m_1 + m_2}$.

Final KE: $K_f = \frac{1}{2}(m_1 + m_2)V^2 = \frac{1}{2} \frac{m_1^2 u_1^2}{m_1 + m_2}$.

Fraction of initial KE retained:

$$\frac{K_f}{K_i} = \frac{\frac{1}{2} \frac{m_1^2 u_1^2}{m_1 + m_2}}{\frac{1}{2} m_1 u_1^2} = \frac{m_1}{m_1 + m_2}$$

Sol 14. Option (A)

For a collision against a stationary surface:

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} = \frac{v'}{v}$$

Sol 15. Option (C)

Velocity of approach:

$$u_{\text{rel}} = u_1 - u_2 = 4 - (-2) = 6 \text{ m/s}$$

Velocity of separation:

$$v_{\text{rel}} = e \cdot u_{\text{rel}} = 0.5 \times 6 = 3 \text{ m/s}$$

Sol 16. Option (A)

Using velocity formula with $M \gg m$ and $u_2 = 0$:

$$v_2 = \left(\frac{M(1+e)}{M+m} \right) V \approx (1+e)V$$

Sol 17. Option (A)

Speed of pendulum block after bullet emerges:

$$V = \sqrt{2gh}$$

By conservation of momentum:

$$mv = m\left(\frac{v}{3}\right) + MV \Rightarrow \frac{2}{3}mv = M\sqrt{2gh} \Rightarrow v = \frac{3M}{2m}\sqrt{2gh}$$

Sol 18. Option (B)

Parallel to floor: momentum is conserved ($u \sin \alpha = v' \sin \beta$).

Perpendicular to floor: $v' \cos \beta = e(u \cos \alpha)$.

Dividing both:

$$\frac{\sin \beta}{\cos \beta} = \frac{u \sin \alpha}{eu \cos \alpha} \Rightarrow \tan \beta = \frac{\tan \alpha}{e}$$

Sol 19. Option (A)

When string tightens, the component of velocity along the string ($u/\sqrt{2}$ at 45°) is destroyed inelastically:

$$\Delta K = \frac{1}{2} \mu u_{\parallel}^2 = \frac{1}{2} \left(\frac{m}{2} \right) \left(\frac{u}{\sqrt{2}} \right)^2 = \frac{1}{4} m u^2$$

Sol 20. Option (A)

The average speed depends directly on the number of bounces n and the restitution coefficient e through the geometric sum of total distance and total time.